

Analysis group meeting

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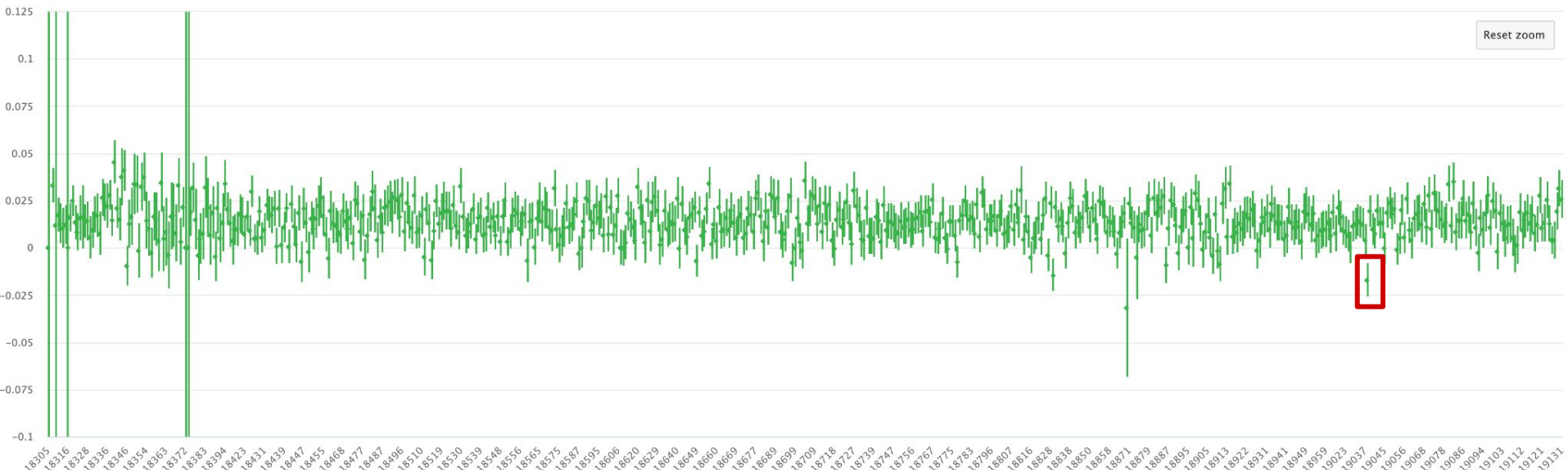
- High level physics QA timelines are published (pass0v10)
- Vz mean timelines show now the two means

- Copy skim runs to cache/hallb/scratch/rg-d/production/skim_pass0v10/

High level physics QA timelines

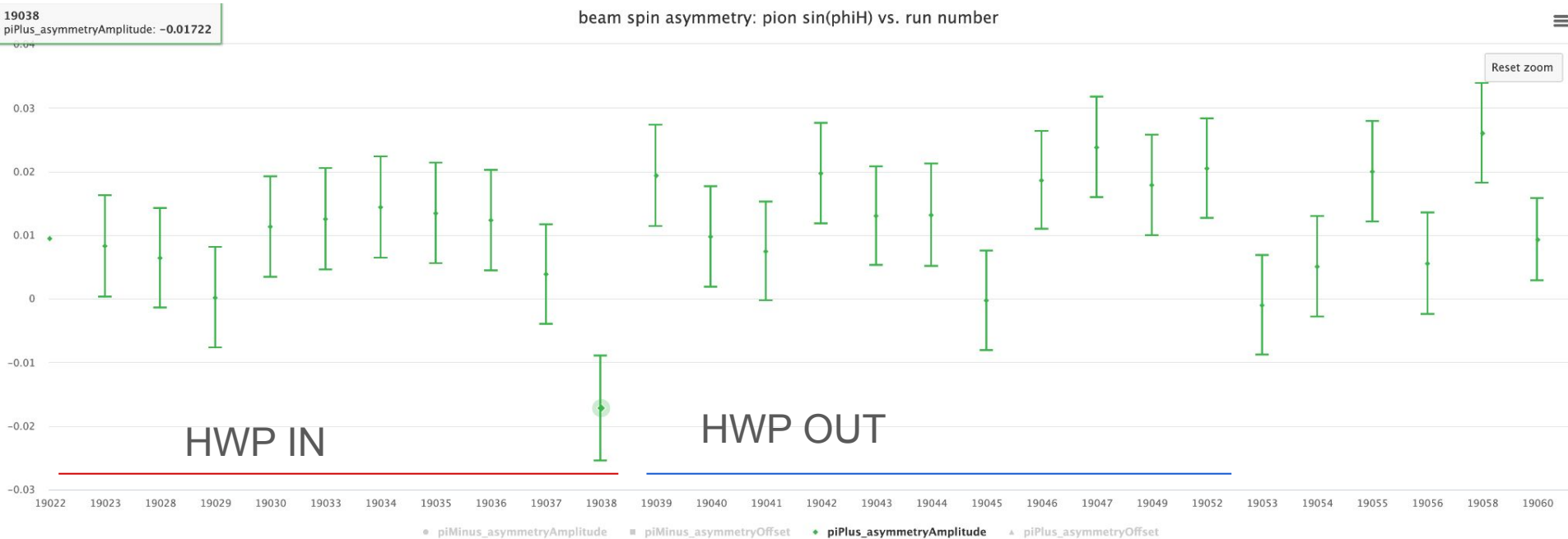
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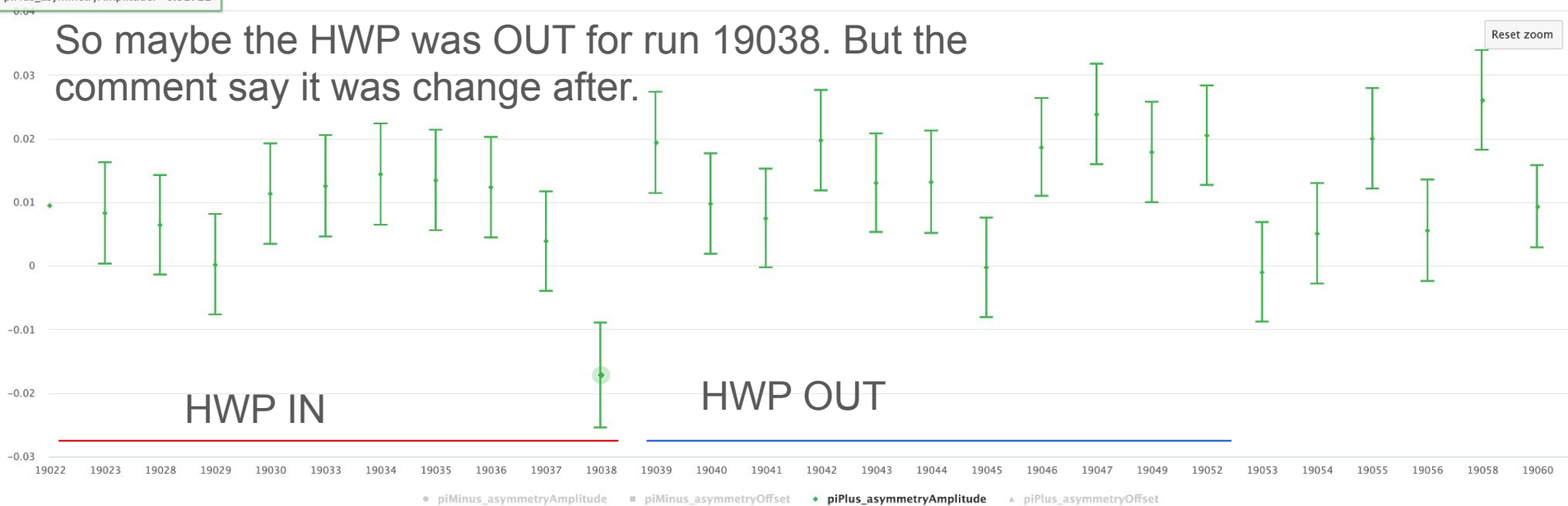
High level physics QA timelines

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19038
piPlus_asymmetryAmplitude: -0.01722

beam spin asymmetry: pion sin(phiH) vs. run number

So maybe the HWP was OUT for run 19038. But the comment say it was change after.



High level physics QA timelines

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8:00 start shift with 19036 25M

8:05 BMT HV SEC2 L4 trip. Turned ON.

8:20 BMT HV SEC2 L4 trip. Turned ON.

9:20 MVT HV SEC2 L5 OFF/ON..

9:24 Stop run 19036 with 101M, start 19037

11:10 BMT HV SEC2 L5 OFF/ON..

11:17 Stop run 19037 with 100M events, start 19038

11:30 BMT HV SEC2 L4 OFF/ON.

11:55 Stop run 19038 with 33M, changing the HWP to OUT (Hall-C request).

12:00 sec-5 tof/pcal/ecal rocs were rebooted, trying to fix problems with sector 5 observed in monitoring of TOF and EC.

12:06 Start run 19039

1:45 DAQ alarm, Sec5 not counting. Stop run 19039 with 96M. Sergei suggested to reboot again the Sec5 ecal and pcal adc rocs.

1:48 Start run 19040

3:33 Stop Run 19040 with 103M events. End of Run failed. Sergei will work on the DAQ.

3:35 Start run 19041

- Need bank `TimeBasedTrkg::TBClusters` and `TimeBasedTrkg::TBTracks`
- Only available with `dcalign` schema.
- Need around 200 files to train the network (wiki say 100 but Raffaella say more)
- Need to use DENOISING for the cooking
- Should we use a mixture of targets or just LD2 for the training?
 - Need to test, train two network one on LD2 the other on CuSn
 - Test them on both LD2 and CuSn and compare the performance
 - If they are similar on both data we can use only LD2 if not we need to train a mixture
- We should also compare the networks train by Daniel, the new one and the RG-B one.
- For the training we can use any good runs that represent the data (if different current are available use the higher one)

- Most of our data are outbending.
- But we validate the network on inbending data (at least for LD2)

Run Number	#18318	#18319	#18321	#18324	#18325	#18326
Luminosity	5nA	20nA	35nA	50nA	75nA	100nA

- Raffaella told me that we should train the network using an outbending run.
- It should be fine to validate it on inbending runs

- Work on a workflow to be able to easily reproduce the MC vs data efficiency plots.
- To fully finish that I need to test the data part:
 - I need to cook some files in the workflow.
 - I will run that on my own account, and cook only 5 events per runs
- I am working on the Brief Progress Report for 22 GeV Frascati Workshop
- I asked Raffaella about the `-LUMI_EVENT="10000, 120*ns, 2*ns"` command.
- It is not important for standard (not lumi) simulation but better 2.004ns
- For the shift in V_z between inbending and outbending, she don't exactly know the reason, we should use two set of cuts.
- The shift is different for positively and negatively charged particle.

- RG-D analysis:
 - AI network + validate
 - V_z cuts for pass0v10 + V_z vs. θ
 - Redo the background merging studies
 - improve pid, fiducials cuts
 - Understand the lc vs TA dependence
 - Finish the event mixing method (difference at low invariant mass)
 - Extract the multiplicity ratio, pT broadening and asymmetry
- ALERT:
 - Nathan give me some comments about my pull request to improve it.
 - Raphael ask me to check my track finder on the data, and add a cut on ADC to remove hits with low ADC.